

gdTAI V4 Step 1 Preflight and Split Freeze

- gdTAI V4 Step 1 Preflight and Split Freeze
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gdTAI V4 Step 1 Preflight and Split Freeze

Overall status: FAIL
Protocol: v1.1
Training started: No
Step 2 authorized: No; supervision approval is still required

Decision

This workflow performed only the precommitted read-only input, truth-label, feature, exclusion-feasibility, and grouped-split checks. It did not fit, calibrate, threshold, compare, or promote a classifier.

Failures block Step 2. Warnings require explicit supervision. The frozen CD4/Treg thresholds were evaluated exactly as written and were not tuned.

Blocking failures

check_id	observed	required	details
training_feature_coverage	gdt_2020aug_wocov;maltegdt	>=0.9 and all critical genes	Hard-gated development inputs
raw_count_state	hra005041	all sparse values finite, nonnegative, integer-like	Full sparse data-array scan

Warnings

No rows.

All preflight checks

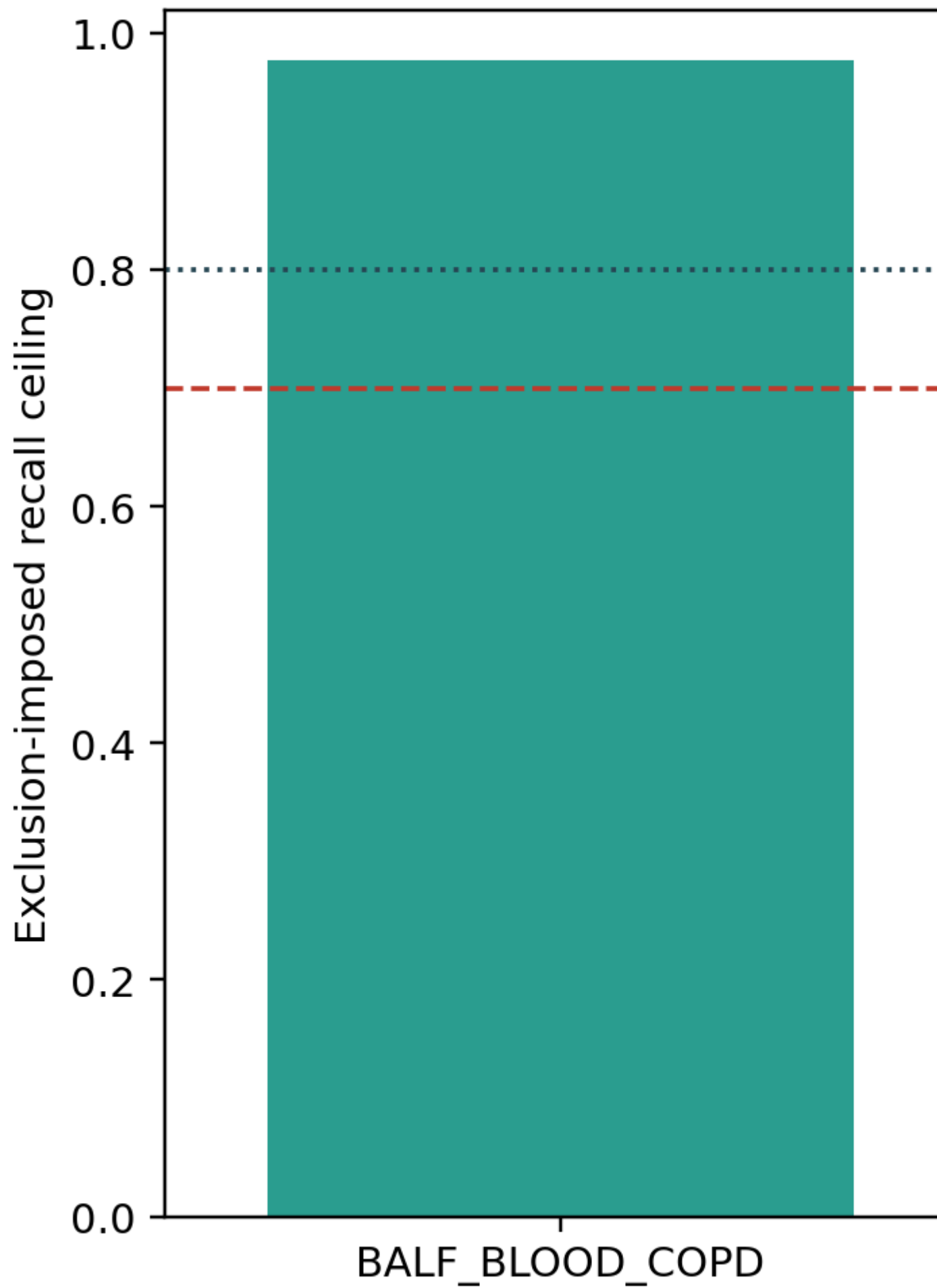
check_id	status	observed	required	details
protocol_version	PASS	1.1	1.1	Frozen protocol version

check_id	status	observed	required	details
feature_count	PASS	197	197	Frozen individual-gene universe
training_feature_coverage	FAIL	gdt_2020aug_wocov;malte gdt	>=0.9 and all critical genes	Hard-gated development inputs
raw_count_state	FAIL	hra005041	all sparse values finite, nonnegative, integer-like	Full sparse data-array scan
registered_hashes	PASS		all available expected hashes match	Full SHA-256 unless explicitly skipped
input_file_state	PASS	16	16	Size and mtime unchanged after read-only workflow
gse144469_join	PASS	107068	107068 unique SRR + barcode mappings	Raw expression to canonical metadata
primary_labels_nonempty	PASS	{'abT_primary': {'BALF_BLOOD_COPD': 32404, 'GSE144469': 60175, 'HRA005041': 543028}, 'gdT_primary': {'BALF_BLOOD_COPD': 1033, 'GSE144469': 4003, 'HRA005041': 4894}}	both classes in each primary source	Expression-independent productive TCR rules
primary_label_overlap	PASS	0	0	Mutually exclusive primary labels
silver_excluded_from_training	PASS	0	0	Sensitivity-only silver cells
cd4_treg_recall_ceiling	PASS	0.983805	macro >= 0.8; each source >= 0.7	Fixed post-model exclusions before Step 2
group_leakage	PASS	1	1	No group appears in multiple inner folds
outer_fold_nonempty	PASS	1	True	Every stage has train and held-out cells
nk_stratum_composition	PASS	{'representative': 0.6, 'gdt_like_TRDCpos_TRDV_neg': 0.2, 't_like_hard': 0.2}	{'representative_fraction': 0.6, 't_like_fraction': 0.2, 'gdt_like_fraction': 0.2}	All strict NK negatives assigned once
supplemental_nk_pool	PASS	323771	>0	Mapped, no-productive-TCR atlas NK negatives
legacy_v4_archived	PASS	True	True	No overwrite of the earlier experimental artifact

CD4/Treg recall cost

The recall ceiling is $1 - \text{union_excluded} / \text{gdT_primary}$. It is the maximum post-exclusion recall even for a perfect pre-exclusion classifier.

Fixed CD4/Treg



Recall ceiling

source_gse_id	n_cells	cd4_helper_only	treg_only	overlap	union_excluded	recall_ceiling	applicable_floor	margin_above_floor
BALF_BLOOD_COPD	1033	22	1	0	23	0.9777	0.7000	0.2777
GSE144469	4003	65	6	18	89	0.9778	0.7000	0.2778
HRA005041	4894	20	0	0	20	0.9959	0.7000	0.2959
SOURCE_MACRO	9930	107	7	18	132	0.9838	0.8000	0.1838

Frozen feature coverage



Feature coverage

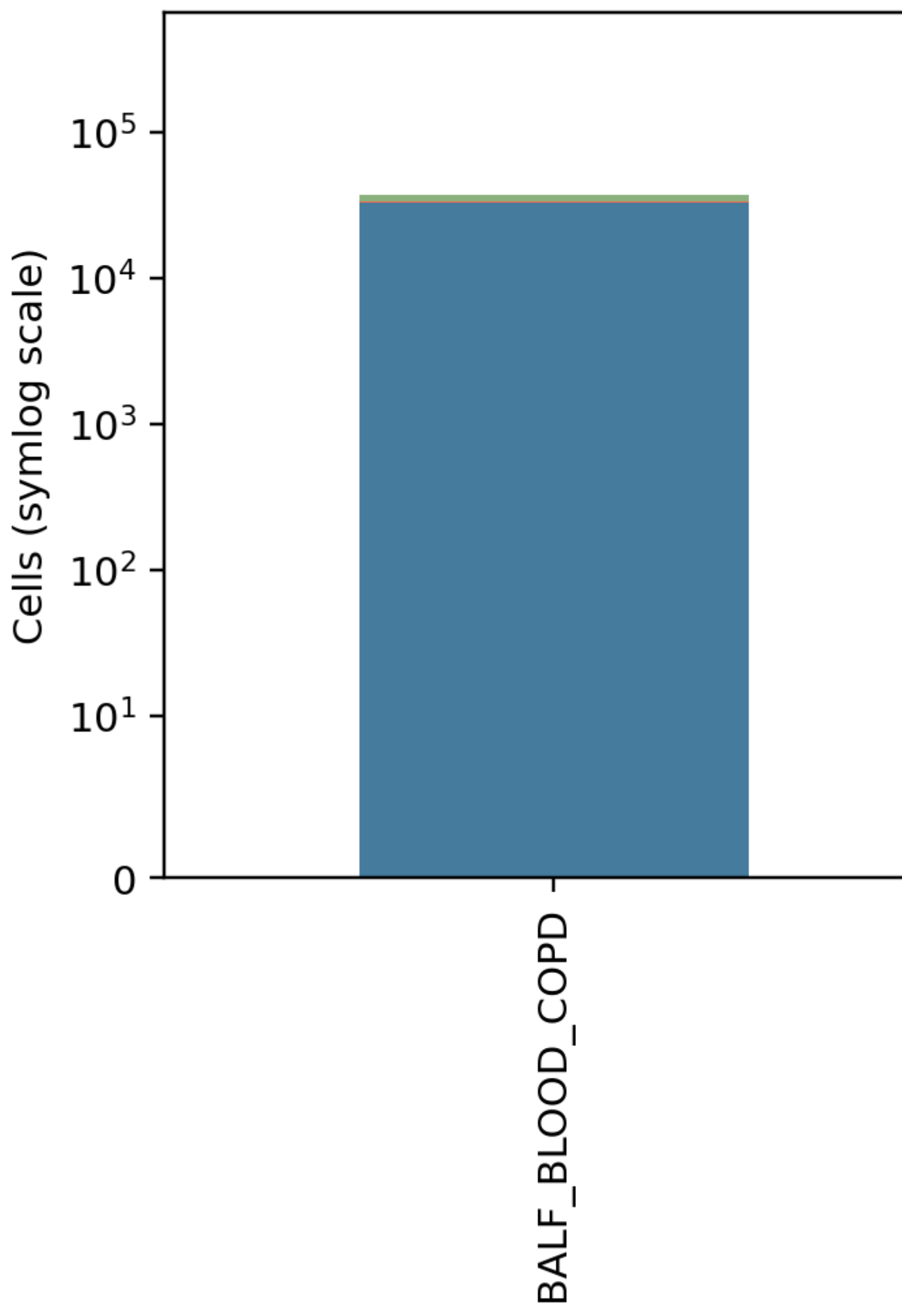
input_id	role	features_present	features_expected	feature_coverage	missing_critical_count	hard_feature_gate
current_atlas	stage1_NK_expression_and_future_atlas_inference	197	197	1.0000	0	True
hra005041	primary_development	197	197	1.0000	0	True
gse144469	primary_development	197	197	1.0000	0	True
half_blood_copd	primary_development_reused_benchmark	197	197	1.0000	0	True
gdt_2020aug_wocov	sorted_training_supplement	163	197	0.8274	0	True
maltegdt	sorted_training_supplement	82	197	0.4162	0	True
gdtlung	sorted_sensitivity_only	197	197	1.0000	0	False
extension_GSE114724	frozen_negative_stress	197	197	1.0000	0	False
extension_GSE121636_GSE121637	frozen_negative_stress	197	197	1.0000	0	False
extension_GSE159251	frozen_negative_stress	188	197	0.9543	0	False
extension_GSE169246	reduced_feature_schema_sensitivity	145	197	0.7360	0	False
extension_GSE292700	frozen_negative_stress	197	197	1.0000	0	False
extension_GSE294273_GSE294274	frozen_negative_stress	197	197	1.0000	0	False
extension_GSE296954	frozen_negative_stress	197	197	1.0000	0	False
extension_GSE315928	frozen_negative_stress	197	197	1.0000	0	False

Raw-count audit

Every stored sparse value was scanned. Integer-like means its distance from the nearest integer was no greater than $1e-6$.

The HRA005041 matrix is known **log1p(CP10K)** and is used here only to quantify exclusion cost. It remains ineligible for fitting because no raw-count layer is available.

input_id	matrix_key	configured_matrix_state	n_obs	nnz	dtype	fractional_values_gt_1e_6	raw_count_pass
current_atlas	X	raw_counts	3705306	5503996785	float32	0	True
hra005041	X	log1p_cp10k_recall_audit_only	766639	1442826937	float64	1442826905	False
gse144469	layers/counts	raw_counts	107068	215368848	float32	0	True
half_blood_copd	layers/counts	raw_counts	46273	125599864	float32	0	True
gdt_2020aug_wocov	X	raw_counts	25904	39050646	int64	0	True
maltegmt	X	raw_counts	7800	12020698	int64	0	True
gdtlung	X	raw_counts	15175	15468404	int64	0	True
extension_GS E114724	X	raw_counts	28652	50878640	int64	0	True
extension_GS E121636_GSE121637	X	raw_counts	18793	28893248	int64	0	True
extension_GS E159251	X	raw_counts	68898	81434083	int64	0	True
extension_GS E169246	X	raw_counts	354373	430559363	int64	0	True
extension_GS E292700	X	raw_counts	78254	127198003	int64	0	True
extension_GS E294273_GSE294274	X	raw_counts	55744	111603853	int64	0	True
extension_GS E296954	X	raw_counts	86608	122094020	float32	0	True
extension_GS E315928	X	raw_counts	66813	91568523	float32	0	True



Label composition

source_gse_id	truth_class	n_cells
BALF_BLOOD_COPD	abT_primary	32404
BALF_BLOOD_COPD	dual_or_ambiguous	2938
BALF_BLOOD_COPD	gdT_primary	1033
BALF_BLOOD_COPD	gdT_silver	31
BALF_BLOOD_COPD	single_abT_weak	3923
BALF_BLOOD_COPD	unlabeled	280
GDT_2020AUG_woCOV	sorted_gdT_weak	25904
GDTlung2023july_7p	sorted_sensitivity	15175
GSE125527	unlabeled	3797
GSE144469	abT_primary	60175
GSE144469	dual_or_ambiguous	5857
GSE144469	gdT_primary	4003
GSE144469	gdT_silver	415
GSE144469	single_abT_weak	3379
GSE144469	unlabeled	8662
GSE145926	unlabeled	361
GSE162498	unlabeled	129
GSE171037	unlabeled	225
GSE178882	unlabeled	5493
GSE188620	unlabeled	20162
GSE190870	unlabeled	1
GSE211504	unlabeled	303
GSE212217	unlabeled	25627
GSE228597	unlabeled	17507
GSE234069	unlabeled	136
GSE235863	unlabeled	18132
GSE240865	unlabeled	364
GSE243013	unlabeled	42490
GSE243572	unlabeled	17075
GSE243905	unlabeled	18659

GSE144469 join

source_gse_id	source_cells	source_unique_keys	canonical_cells	canonical_unique_keys	mapped_cells	unmapped_cells	mapping_fraction	source_duplicate_keys	canonical_duplicate_keys	obs_sample_disagrees_with_id_srr	obs_sample_barcode_unique_keys
GSE144469	107068	107068	107068	107068	107068	0	1.0000	0	0	16851	90217

Frozen outer folds

outer_fold_id	heldout_source	stage	outer_train_cells	outer_train_positive	outer_eval_cells	outer_eval_positive	outer_train_groups
outer_0_HRA005041	HRA005041	stage1	486115	154286	641989	637038	751
outer_0_HRA005041	HRA005041	stage2	138621	38740	627813	4894	31
outer_1_GSE144469	GSE144469	stage1	1045613	716637	82491	74687	732
outer_1_GSE144469	GSE144469	stage2	698877	39631	67557	4003	28
outer_2_BALF_BLOOD_COPD	BALF_BLOOD_COPD	stage1	1087495	750969	40609	40355	764
outer_2_BALF_BLOOD_COPD	BALF_BLOOD_COPD	stage2	729074	42601	37360	1033	44

Input identity

input_id	role	size_bytes	sha256	expected_hash_match
current_atlas	stage1_NK_expression_and_future_atlas_inference	73350311676	9060920d6fe19bdb	True
legacy_metadata_reference	read_only_TCR_and_GSE144469_metadata_reference	24383355208	f44bd4941a7b8c8d	True
hra005041	primary_development	19626113528	fc9ead68c3d85232	True
gse144469	primary_development	1536829723	456b2da064c9ad60	True
balf_blood_copd	primary_development_reused_benchmark	2110825599	acf33fcc3851f77c	True
gdt_2020aug_wocov	sorted_training_supplement	591495928	650f583b85f05ede	True
maltegdt	sorted_training_supplement	179862512	bbd460ef153ce6b2	True
gdtlung	sorted_sensitivity_only	276869875	d5915e6b50241ff8	True

input_id	role	size_bytes	sha256	expected_hash_match
extension_GSE114724	frozen_negative_stress	131909269	394425c7937432ac	True
extension_GSE121636_GSE121637	frozen_negative_stress	77281297	b7182274cab834f1	True
extension_GSE159251	frozen_negative_stress	223142214	abe6502b3a15a3ba	True
extension_GSE169246	reduced_feature_schema_sensitivity	2176683186	f251382c62109a48	True
extension_GSE292700	frozen_negative_stress	326784213	78c9ada980a21592	True
extension_GSE294273_GSE294274	frozen_negative_stress	289243072	6a20d61544c7cd33	True
extension_GSE296954	frozen_negative_stress	304970784	f3484482e35b49d3	True
extension_GSE315928	frozen_negative_stress	253771849	2fe674761db67aa3	True

Canonical artifacts

- Cell-label manifest: `Integrated_dataset/tables/gdT_prediction/gdtai_v4_preflight/cell_label_manifest.csv.gz`
- Inner group assignments: `Integrated_dataset/tables/gdT_prediction/gdtai_v4_preflight/inner_group_split_manifest.csv.gz`
- Full checks: `Integrated_dataset/tables/gdT_prediction/gdtai_v4_preflight/preflight_checks.csv`
- Machine summary: `Integrated_dataset/logs/gdT_prediction/gdtai_v4_preflight/preflight_summary.json`
- Reader HTML: `gdT_prediction/gdtai_v4_preflight/index.html`
- PDF: `gdT_prediction/gdtai_v4_preflight/gdtai_v4_preflight_report.pdf`

Gate

Stop before Step 2. No model fitting is permitted until the user reviews this package and explicitly approves the next gate.